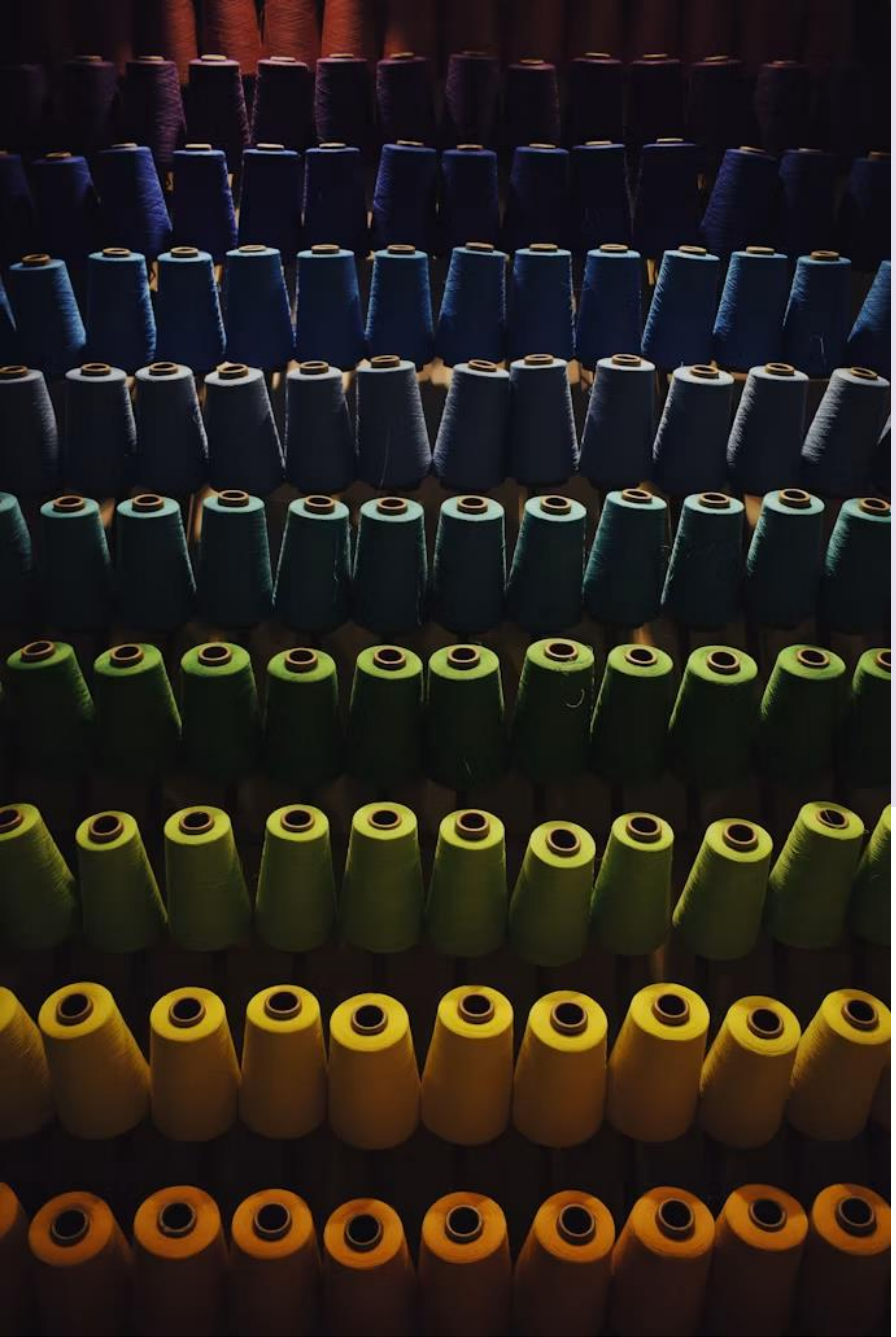

FABRIC MANUFACTURING

Part: B

Course Code: FE 253-0723

Mechanical Principles of Knitting Technology

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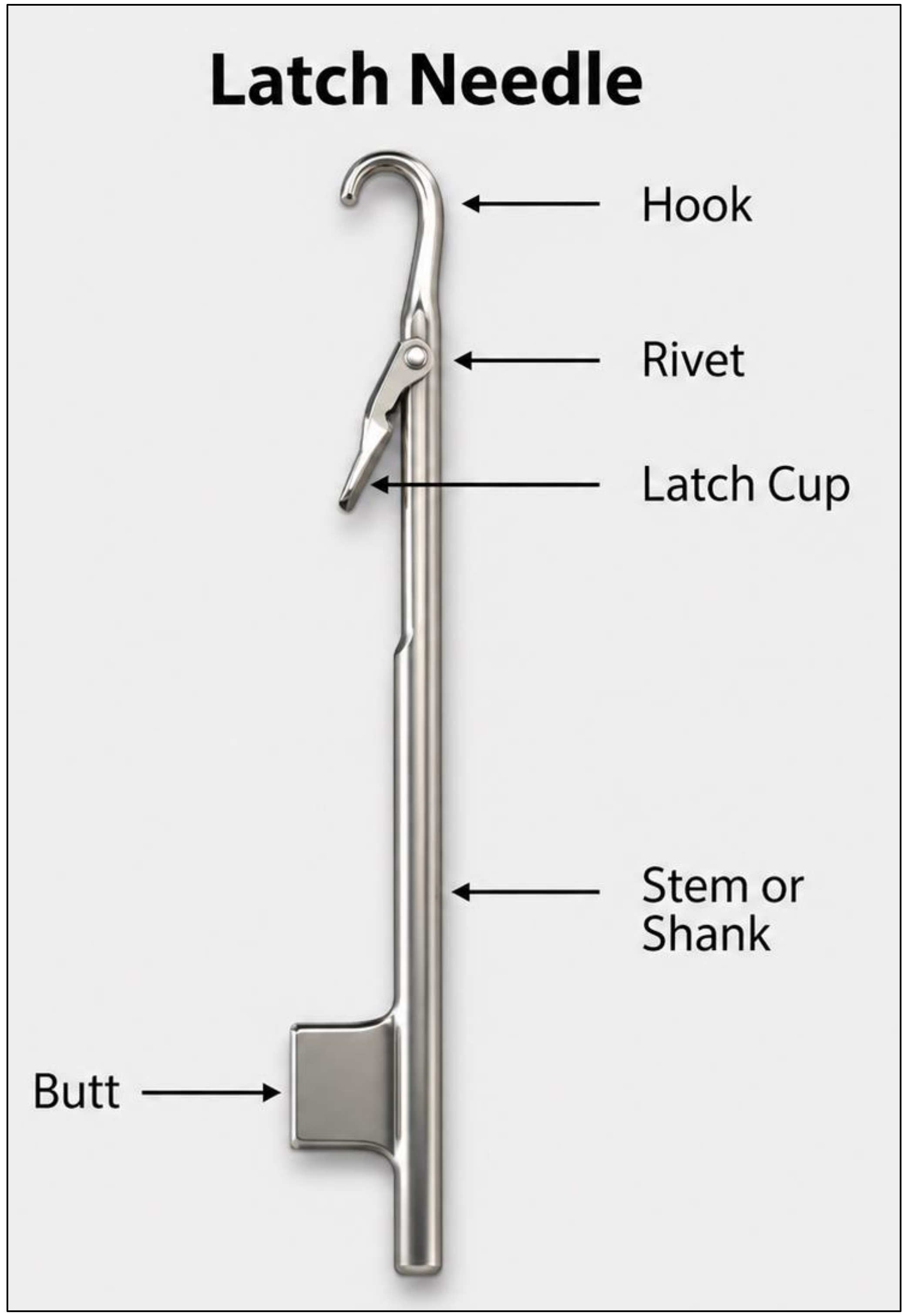
BASIC ELEMENTS

<u>Needle</u>

The primary knitting element in knitting machine is needle which makes a compound movement for producing loops. In circular weft knitting, the needle moves round the machine axis along with the cylinder and at the same time performs an upward and downward movement. But in flat-bed knitting machine, needle is fixed onto the needle bed. Needles are mainly of three types.

1. Latch needle
2. Spring bearded needle
3. Compound needle.



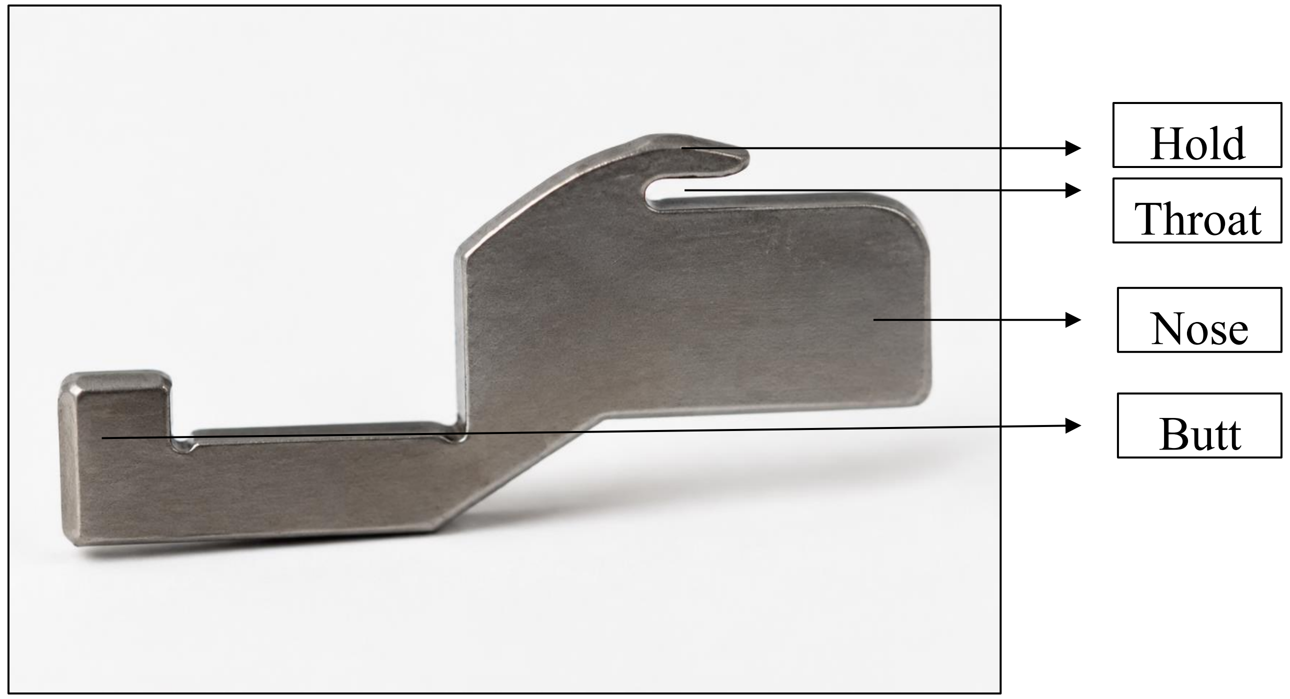
BASIC ELEMENTS

Sinker

Sinker is the second primary knitting element though double jersey knitting machines do not use sinker. It is a specially shaped flat metal plate that sits in between each needle in single jersey knitting machine and warp knitting machine. The sinker helps in loop formation, holding down and knocking over (the needle passes through the old loop by completely drawing a new loop). Sinkers are placed horizontally.

It is mainly of three types:

1. Holding down sinker (CKM),
2. Knocking over (Warp knitting)
3. Loop Forming sinker (Bearded needle-based weft knitting machine straight bar frame)



BASIC ELEMENTS

Cam:

Cam is the third primary knitting elements which convert the rotary machine drive into a suitable reciprocating action to the needles or other elements. Cam assists the needle in loop formation. Knitting cams are of three types:

1. Knit cam
2. Tuck cam
3. Miss cam

Knit Cam　Tuck Cam　Miss Cam

NEEDLES

Spring Bearded Needle

The spring-bearded needles made of steel wire consist of the following parts:

a. The Stem: The stem around which the needle loop is formed.
b. The Head: Where the stem is turned into a hook to draw the new loop through the old loop.
c. The Beard: Which is the curved downwards continuation of the hook that is used to separate the trapped new loop inside from the old loop as it slides off the needle beard.
d. The Eye or groove: Cut in the stem to receive the pointed tip of the beard when it is pressed, thus enclosing the new loop.
e. Shank: Serving to attach the needle in the needle bar.

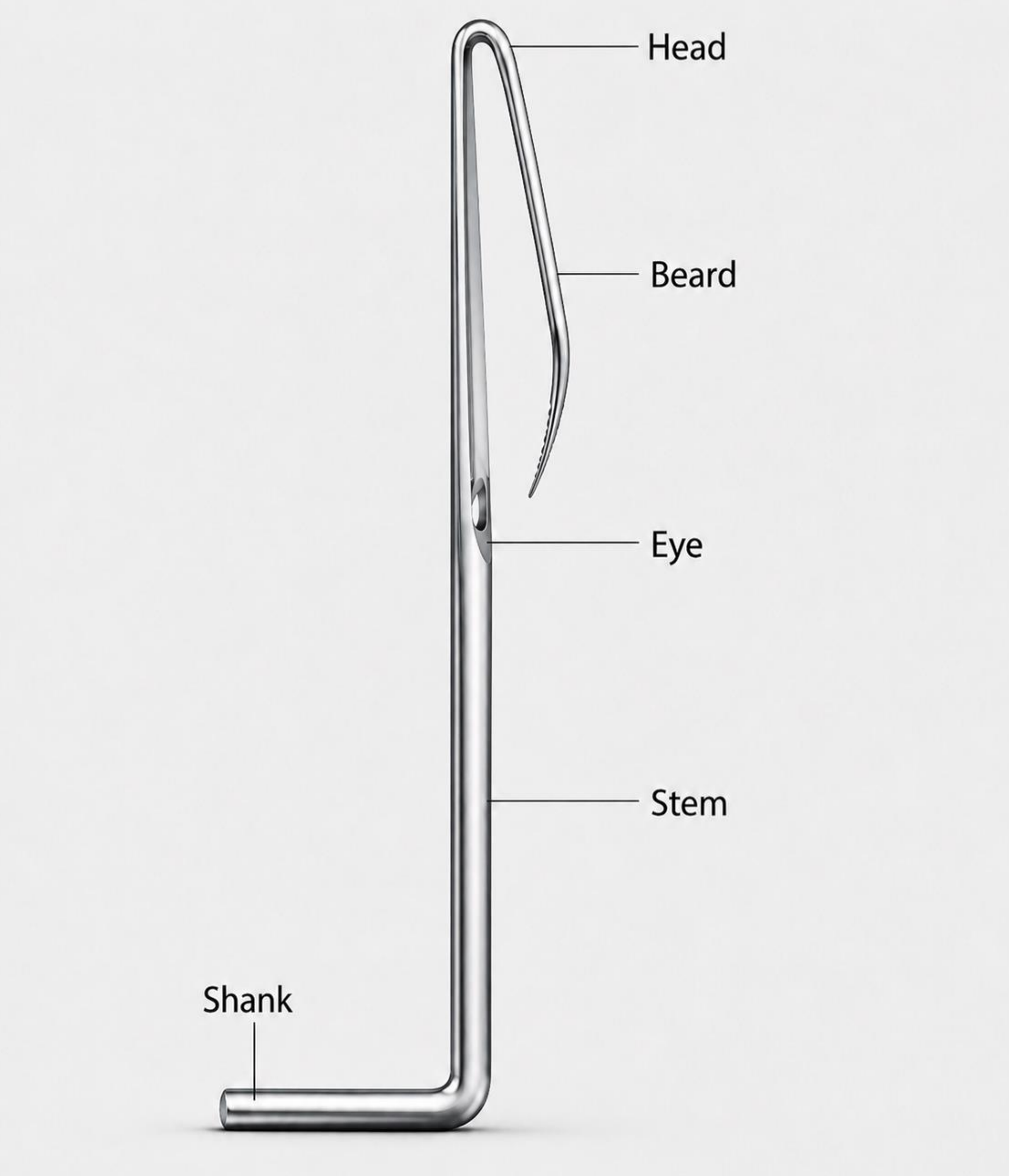


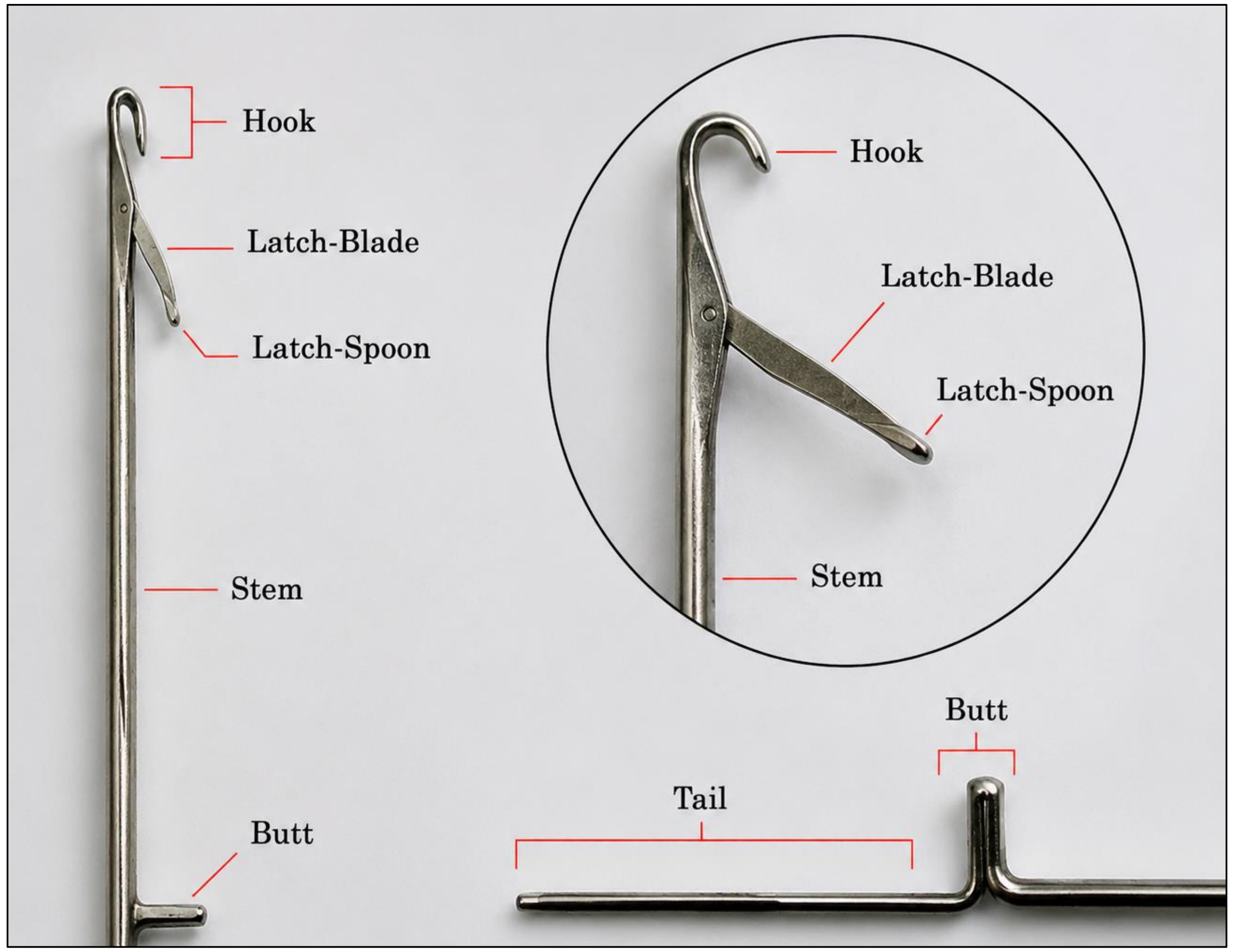
Fig: Spring Bearded needle

NEEDLES

Latch Needle

The latch needles have following parts:

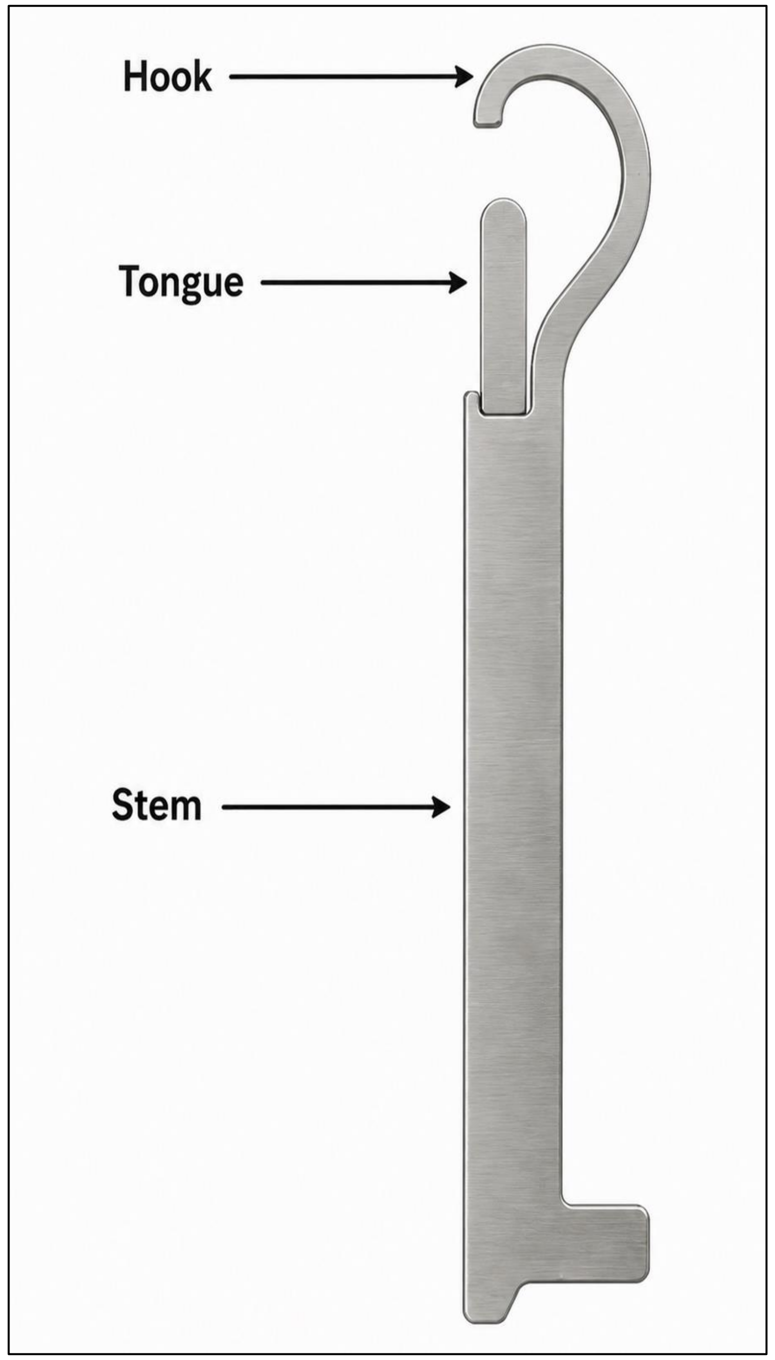
1. Hook: Grasping a new yarn in the knitting process.
2. Latch: Freely rotating around the axle.
3. Latch Blade: Locates the latch in the needle.
4. Latch Spoon: Extension of latch blade bridges the gap between hook when closed.
5. Stem: Carries the loop in clearing position.
6. Rivet: This has been dispensed with on most plate metal needles by pinching in the slot walls to retain the latch blade.
7. Butt: Which serving to displace the needle along the needle bed slot.
8. Tail: Extension below the butt giving additional support.



NEEDLES

Compound Needle

- Compound Needle consists of two separately controlled parts; these are- the open hook and the hook closing element (tongue, latch, piston, and plunger).
- The two parts rise and fall as a single unit but at the top of the rise, the hook moves faster to open and at the start of the fall the hook descends faster to close the hook.
- Knitting action is short, smooth, and simple. So, the production rate is the highest for this needle. Also, yarn is subjected to lowest strain.
- It is easier to drive the hooks and tongues collectively from two separate bars as in warp knitting since no presser or rivet are used like bearded and latch needle respectively. It is widely used in warp knitting due to the leverage of attaining higher knitting speed.
- Manufacturing cost is higher.



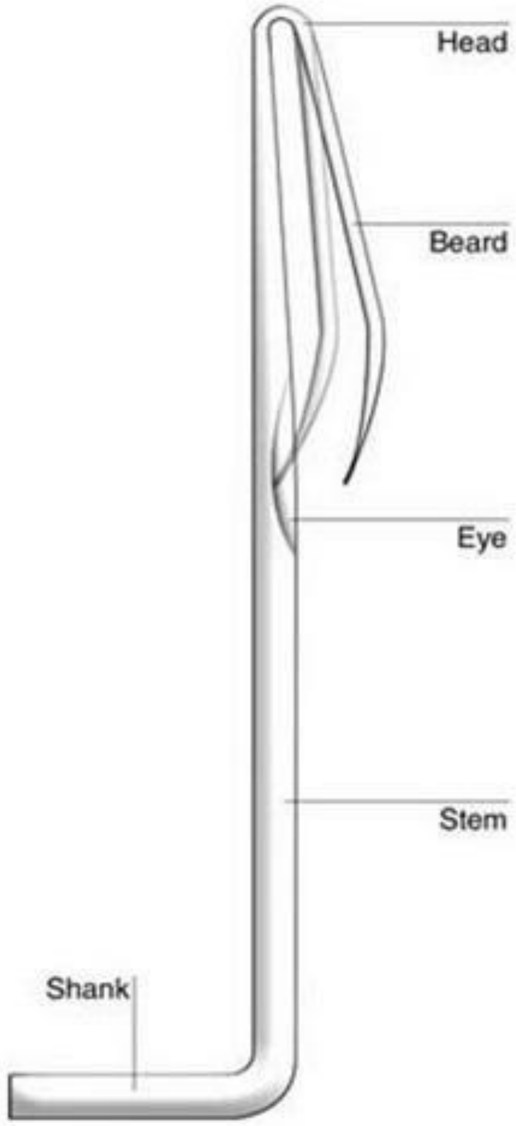
DIFFERENT TYPES OF NEEDLE

Latch Needle for CKM

Latch needle for FKM

DE Latch Needle for PKM

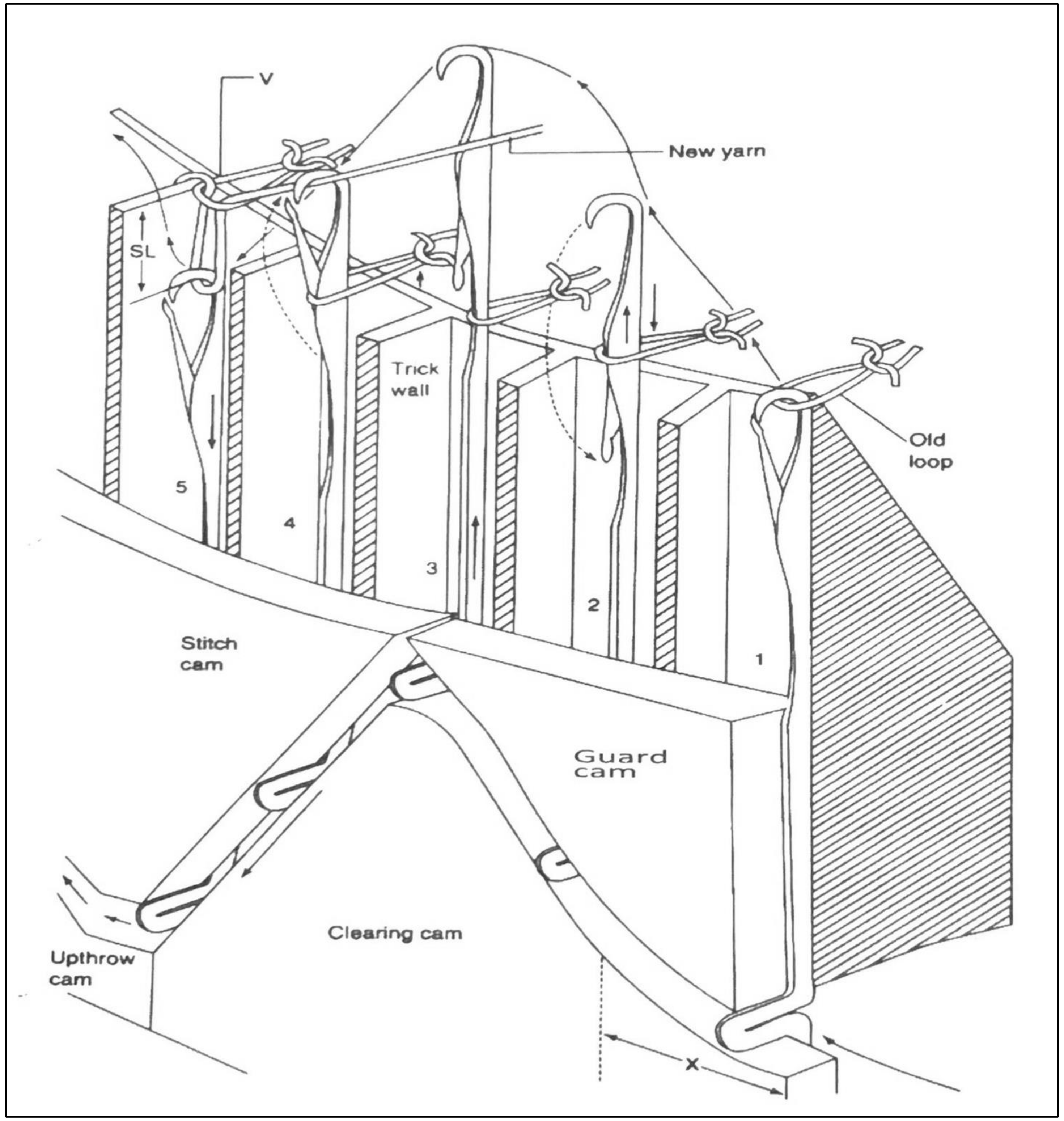
Latch Needle for Warp KM



Bearded Needle for Warp KM

Compound Needle for Warp KM

PARTS OF A KNIT CAM



1. The Clearing Cam: This causes the needles to be lifted to either tuck, clearing loop transfer or needle transfer height depending upon machine design. Clearing/Raising cam is mainly classified into knit, tuck, and miss.

2. The Stitch Cam: It controls the depth to which the needle descends thus controlling the amount of yarn drawn into the needle Ioop.

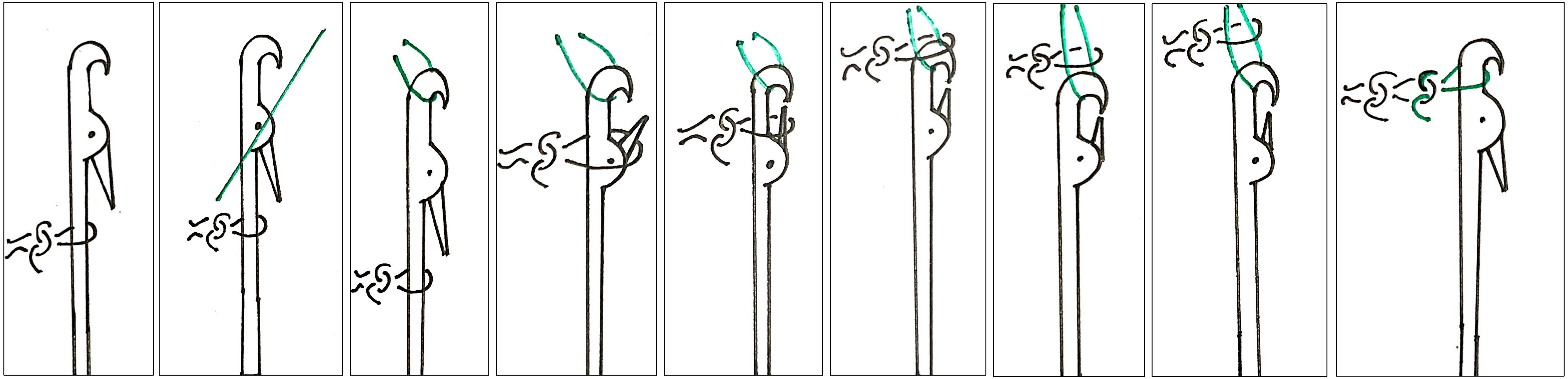
3. The Up-throw Cam: It takes the needles back to the rest position and allows the newly formed loops to relax.

4. The Guard Cam: These are often placed on the opposite side of the cam race to limit the movement of the butts and to prevent needles from falling out of track.

KNITTING CYCLE OF LATCH NEEDLE

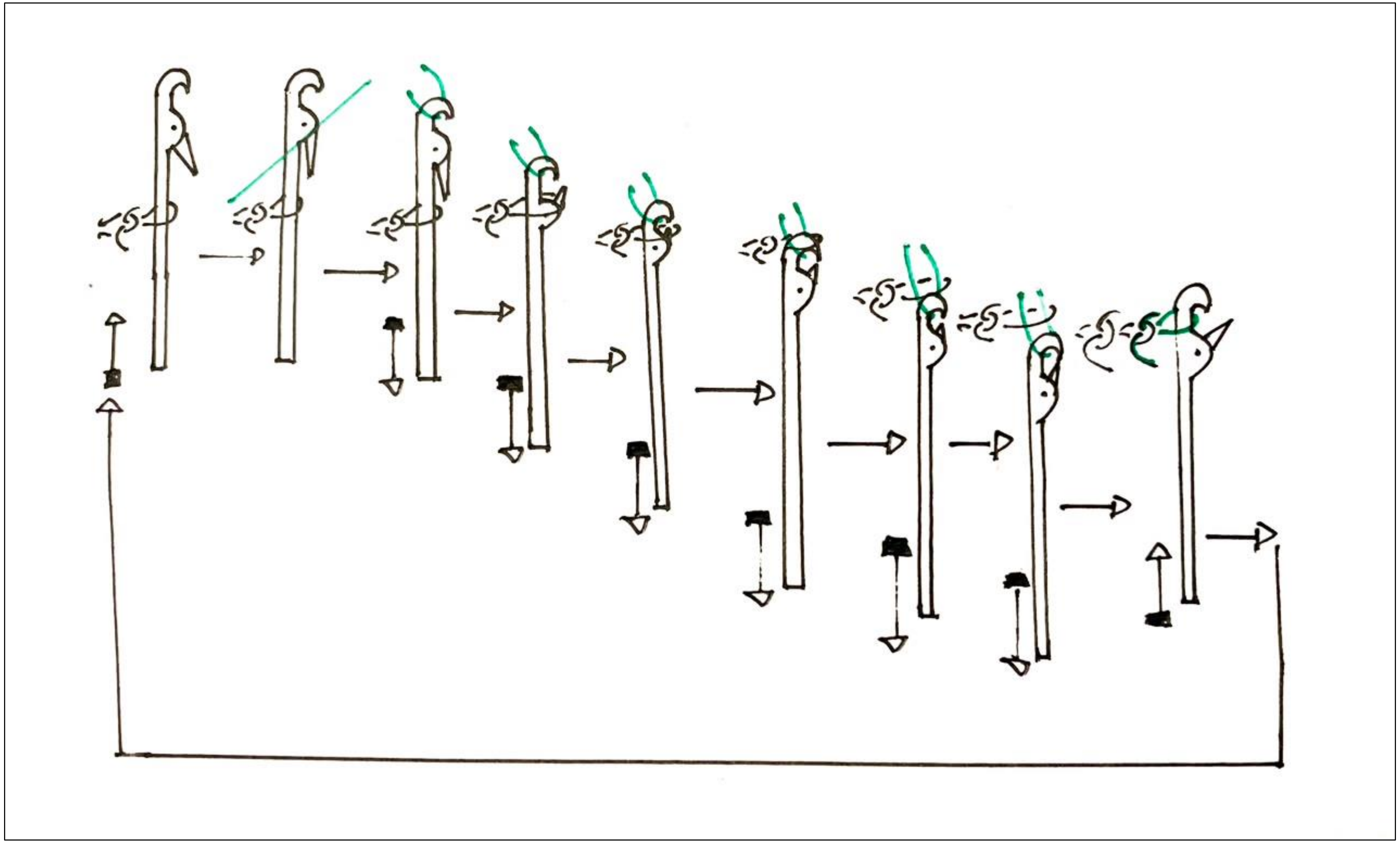
1. Clearing: The process of stitch formation is started by the clearing operation. Its aim is to draw the old loops behind the needle latch into the needle stem. Clearing is completed when the needle reaches its upper position.
2. Yarn feeding: In the course of needle downward motion, the hook takes hold of yarn coming from the yarn guide.
3. Underlapping: Further movement of yarn just laid and its forwarding under the needle hook is called under lapping.
4. Pressing: The aim of pressing is to close the needle hook with the yarn laid in it.
5. Landing: The essence of this operation consists in shifting the old loop on the closed latch.
6. Joining: At joining, the new yarn comes in contact with the old loop but remain in the exposure of the hook of the needle.
7. Knock over: Old loop completely leaves the needle and intermeshes with the new loop.
8. Stich formation: The needle further descends depending on the depth of the stich cam and a fully completed loop is formed.
9. Loop Draw-off: The aim of this operation is to draw the old loop behind the needle back. It is affected by sinker throats.

KNITTING CYCLE OF LATCH NEEDLE



CLEARING | YARN RECEIVING | UNDER LAPPING | PRESSING | LANDING | JOINING | KNOCKING OVER/CASTING OFF | LOOP FORMING | LOOP DRAW OFF

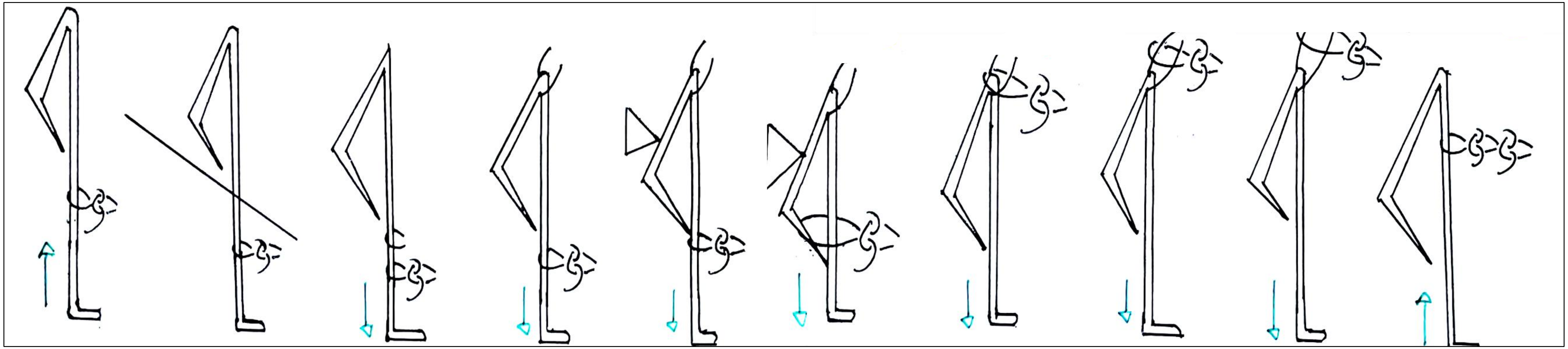
KNITTING CYCLE OF LATCH NEEDLE



KNITTING CYCLE OF SPRING BEARDED NEEDLE

1. Clearing: Needle is placed over the top-most position. The hook is open, and the old loop is placed along the needle stem and again started to form new loop.
2. Yarn feeding: New yarn is feed under the throat of kinking sinker.
3. Yarn Sinking: The new yarn is transformed into loop shape by using throats of loop forming sinker. It is the disadvantage of bearded needle.
4. Underlapping: New loop is formed in the hook.
5. Pressing: Needle beard is pressed in the needle grove by pressure disc.
6. Landing: The old loop lands on the beard. Simultaneously needle continues to move downwards.
7. Joining: The old loop gets closer the new loop as the needle descends but remain in the exposure of the hook of the needle.
8. Casting-off Knocking over: Old loop leaves the needle and intermeshes with the new loop.
9. Stich formation: The needle further descends depending on the depth of the stich cam and a fully completed loop is formed.
10. Loop Draw-off: Due to tension of take down roller, loops become horizontal from vertical position.

KNITTING CYCLE OF SPRING BEARDED NEEDLE



CLEARING

YARN RECEIVING

YARN SINKING

UNDER LAPPING

PRESSING

LANDING

JOINING

KNOCKING OVER/CASTING OVER

LOOP FORMING

LOOP DRAW OFF

THANK YOU !